

FIT3143 Applied #2: Generative AI Declaration

Unit: FIT3143 Parallel Computing, Monash University Malaysia **Assessment:** Applied #2, GPU/NPU and Exponential Growth **Team:** Erwyna Soo Wen Xin (36555789, esoo0013@student.monash.edu) and Taabish Farooq Bhat (35473932, ttaa0006@student.monash.edu) **Date:** 1 October 2026

1. Statement

We used generative AI while preparing this assessment, as the specification allows (instructions 9 and 10). We did not use any AI tool during the presentation or the Q&A. We ran, checked and edited every AI output ourselves. Every measured number in our slides and reports comes from our own run on a Google Colab Tesla T4 ([results/results_8k.csv](#), [results/run_8k.txt](#)), and every cited figure was checked against its source.

2. Tools used

Tool	Provider	Used by	Used for
Gemini	Google	Erwyna and Taabish	Explaining concepts and early research (prompts in section 4)
ChatGPT	OpenAI	Erwyna and Taabish	Explaining concepts and early research (prompts in section 4)

3. What AI produced and how we checked it

Artefact	What AI did	How we checked it
rotate_cuda.cu	Drafted the CUDA program, variants v0 to v6, timing and correctness checks	Syntax check, CPU emulation of kernels, then a full run on a Colab Tesla T4; v1 to v5 match the CPU byte for byte
rotate_cpu.c	Drafted the CPU reference and self-tests	Compiled and run locally, all tests pass
Applied2_Task1_Colab.ipynb	Drafted the notebook	Run end to end on Colab T4; CSVs and graphs g1 to g6 saved
Diagrams D1 to D5, charts C1 to C5	Drafted plotting scripts	Visually checked; every plotted number traced to a cited source or our CSV
Task1_Answers.pdf, Task2_Report.pdf	Produced first drafts	Read, edited and checked every reference and number against sources
Slides and speaker scripts	Drafted text and edited the Canva deck	Rehearsed, timed and edited by both members
Q&A drills	Drafted practice questions	Numbers filled from our own run

Artefact	What AI did	How we checked it
References	Checked DOIs and author lists (Crossref, arXiv) and formatted APA 7 / IEEE	Spot-checked against the source pages

Corrections we made after checking AI and course material: the supplementary lecture's GPUs-per-gigawatt estimate (corrected to about 670,000 H100-class GPUs per GW), the "AI emits more CO2 than aviation" claim (a 2030 projection, not a current fact), and an early draft's "2D blocks win because of the cache" explanation (our block sweep showed the real cause is the 1D divide and modulo).

4. Prompt records

4.1 Gemini and ChatGPT (both members)

Prompt 1: "Explain how CUDA handles host-to-device memory transfers over PCIe, focusing on pageable vs pinned memory and asynchronous streaming."

Prompt 2: "Derive 2D inverse image rotation transformation matrix math and convert it into a CUDA kernel using blockIdx and threadIdx."

Prompt 3: "Evaluate GPUDirect Storage (GDS) for high-resolution image processing pipelines and explain scenarios where GDS is beneficial vs non-beneficial."

Prompt 4: "Discuss ethical implications of scaling HPC supercomputing for AI, including carbon emissions, the compute divide, and Green AI scheduling frameworks."

Full chat exports from each member are attached as Appendix A (Erwyna) and Appendix B (Taabish).